

Kannada handwritten Character Recognition using a hybrid deep Learning approach

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Abstract: *In this research work, we propose a novel approach for recognizing handwritten text using transfer learning. Our method leverages a pre-trained model on a large data corpus as the training data and applies it to recognize Kannada characters. Specifically, we employ the ResNet50 model for this task. ResNet50 is a deep convolutional neural network architecture that has shown excellent performance in image recognition tasks. The ResNet50 model comprises several layers, including convolutional, pooling, and fully connected layers. It consists of 50 layers in total, organized into blocks. The blocks in ResNet50 include residual connections, which facilitate the flow of gradients during training and help mitigate the vanishing gradient problem. ResNet50 has 16 convolutional layers, including the layers within the residual blocks, 3 fully connected layers, and an output layer. Our Kannada character set is constructed with a total of 50,773 characters, which are divided into 493 distinct classes. Each class contains a variable number of samples, ranging from 50 to 300 instances. The dataset covers a wide range of variations in writing styles and provides a comprehensive representation of Kannada characters. Through extensive evaluation, we achieved a validation accuracy of 91% after training the ResNet50 model for 20 epochs. The high accuracy demonstrates the effectiveness of our approach in recognizing handwritten Kannada text. Our findings highlight the potential of transfer learning and the ResNet50 architecture in tackling complex character recognition tasks.*

Keywords— *Transfer learning, handwritten text recognition, ResNet50, Kannada characters, deep learning, convolutional neural networks.*

I. INTRODUCTION

Deep learning systems have emerged as a dominant approach in various fields, showcasing remarkable performance in achieving high accuracies. However, these systems heavily rely on large datasets with proper labeling for training. Acquiring such labeled datasets for specific tasks can be time-consuming and expensive. To address this challenge, transfer learning has gained significant attention in recent years. Transfer learning leverages the knowledge learned from successful intelligent systems trained on large labeled datasets and applies it to new systems, reducing the need for extensive labeled data.

Recognizing handwritten Kannada characters poses a formidable and unresolved research challenge. Despite numerous efforts employing machine learning architectures, attaining high accuracy levels remains a persistent obstacle. Kannada, one of the major Dravidian languages spoken in southern India, has a unique set of characters with complex shapes and structures, making their recognition a complex task. In the field of Kannada handwritten character recognition, several researchers have made significant

contributions to developing models and techniques for accurate recognition. Abhishek et al. [1] proposed a deep learning model based on a specific CNN architecture, showcasing its effectiveness in recognizing Kannada characters. Rani et al. [2] introduced a novel model for the recognition of handwritten Kannada characters, leveraging transfer learning from a Devanagari handwritten recognition system. Deekshith and Kanchana [3] took a different approach by proposing a Kannada handwritten character recognition model that utilized hybrid machine learning techniques. Dutta et al. [4] developed a system for Kannada handwritten character recognition using the Multi-Class SVM Method. Fernandes and Rodrigues [5] explored the use of Tesseract and CNN for recognizing Kannada handwritten scripts. Additionally, Dutta et al. [6] employed various machine-learning methods to achieve accurate Kannada character recognition.

The application of CNNs for Kannada character recognition has been widely explored. Sandhya et al. [7] proposed a DenseNet121-based model that effectively recognizes handwritten Kannada characters. Ramesh et al. [8] utilized CNNs for the recognition of hand-written characters. Joe, Savit and Chandrasekaran [9] focused on performing OCR on segmented Kannada characters using CNN. In a different context, Siddiqua et al. [10] used neural networks for the recognition of Kannada characters in scene images. Asha et al. [11] also developed a CNN model for recognizing Kannada characters. While much research has been conducted on handwritten character recognition, there have also been efforts to recognize printed characters. Sridevi et al. [12] explored deep neural networks for recognizing printed Kannada characters. Similarly, Parikshith et al. [13] utilized CNNs and transfer learning for handwritten Devanagari character recognition.

Beyond Kannada characters, researchers have explored the recognition of other scripts using deep learning techniques. Joshi [14] focused on recognizing Gujarati handwritten characters using deep learning methods. Raundale [15] conducted a study on handwritten character recognition, employing various deep learning techniques. Gnanasivam et al. [16] proposed a CNN model for recognizing handwritten Tamil characters under different lighting environments. Additionally, Dessai and Patil [17] used CNNs for recognizing handwritten Devanagari scripts.

Furthermore, several researchers have made contributions to the broader field of character recognition using machine learning techniques. Ansari [18], Nikitha [19], and Vaidya [20] explored approaches to recognizing handwritten text.

Sharma et al. [21] conducted a survey on character recognition using machine learning techniques. The application of transfer learning has also gained attention in character recognition. Ansari [22] and Aneja [23] specifically applied transfer learning to recognize handwritten Devanagari characters. Vijay et al. [24] conducted research on the recognition of Tamil characters using transfer learning algorithms such as AlexNet and VGGNet19.

In conclusion, Kannada handwritten character recognition is a challenging task that has attracted significant research attention. The use of transfer learning, deep learning architectures, preprocessing techniques, and character recognition approaches has contributed to the advancements in this field. The studies discussed in this introduction provide valuable insights into the various techniques and methodologies employed to improve Kannada handwritten character recognition. By leveraging transfer learning, researchers have achieved enhanced recognition accuracy, paving the way for further exploration and investigation in this domain. These works collectively demonstrate the extensive research conducted in the field of character recognition, specifically focusing on Kannada characters. In this study, we aim to contribute to the existing body of knowledge by utilizing transfer learning and applying it to the recognition of Kannada handwritten characters.

The paper follows a structured organization, beginning with Section 2, which introduces the proposed methodology, detailing the approach used in this research. Experimental analysis is described in Section 3, providing insights into the conducted experiments. Finally, Section 4 and Section 5 provide the concluding remarks and summarize the key findings of the study.

II. PROPOSED METHODOLOGY

The proposed methodology aims to tackle the task of recognizing handwritten Kannada characters through the utilization of a transfer learning framework. Overcoming the inherent challenge of capturing the diverse range of handwriting styles inherent in the Kannada script is of utmost importance. In order to overcome this challenge, we employ the technique of transfer learning, which involves adapting a pre-trained model obtained from a large and diverse dataset to a more limited and specialized dataset tailored for Kannada handwritten character recognition. Transfer learning allows us to leverage the knowledge and representations learned by the pre-trained model, which has already been exposed to a vast amount of data, and apply it to the specific task at hand. By doing so, we can capitalize on the generalization capabilities of the pre-trained model and expedite the learning process on the targeted Kannada handwritten character dataset. This approach not only saves computational resources but also addresses the scarcity of annotated data, which is a common challenge in this domain. By fine-tuning the pre-trained model using the Kannada handwritten character dataset, we can achieve enhanced recognition accuracy and overcome the limitations associated with training from scratch.

Within this study, the pivotal steps of feature extraction and classification are conducted employing Resnet50, an advanced deep convolutional neural network comprising 16 layers. Resnet50 has demonstrated remarkable prowess in effectively categorizing images across disparate domains, even with datasets housing a staggering array of up to 1000 categories. Thus, Resnet50 stands as the cornerstone model employed to successfully classify the extensive repertoire of approximately 493 distinct classes of handwritten Kannada characters.

To thoroughly evaluate the efficacy and performance of our proposed methodology, we embark on a comprehensive experimentation and evaluation process. Specifically, we subject the trained Resnet50 model to a separate test dataset, carefully measuring its accuracy, precision, recall, and F1 score. These metrics serve as reliable indicators of the model's ability to precisely recognize and proficiently classify handwritten Kannada characters. The results yielded from this rigorous evaluation provide valuable insights into the feasibility and potential of transfer learning within the specialized domain of handwritten Kannada character recognition. The amalgamation of transfer learning principles and the inherent efficiency of Resnet50 forms the foundation of our methodology, addressing the intricate challenges associated with recognizing handwritten Kannada characters. By leveraging the knowledge embedded within a pre-trained model and fine-tuning it to adapt to a smaller, domain-specific dataset, we aim to achieve outstanding accuracy and robustness in the character recognition process. In the subsequent sections, we delve into a detailed analysis of our experiments, elucidating the methodology's effectiveness and presenting a comprehensive evaluation of its performance.

Figure 1 illustrates the application of transfer learning, where the knowledge gained from a pre-trained deep learning model is utilized to build another system. The comprehensive breakdown of the 493 classes intended for classification can be found in Table 1, while the specifics of the datasets employed are outlined in Table 2.

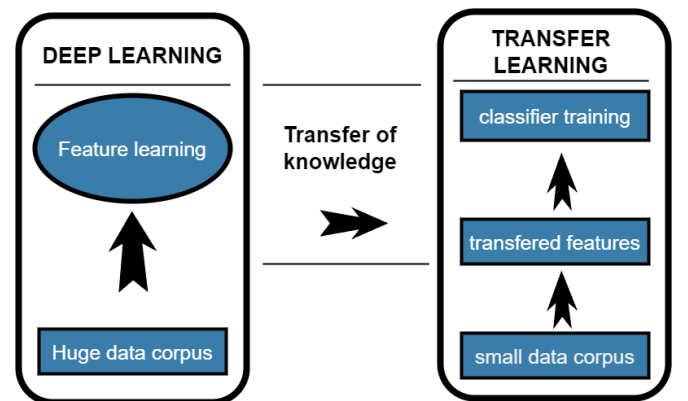


Figure 1: system block diagram

Table 1: Proposed Classification Categories for Kannada Characters

1-ಅ	2-ಆ	3-ಇ	4-ಈ	5-ಉ	6-ಊ	7-ಋ	8-ೠ	9-ಎ	10-ಏ
11-ಐ	12-ಒ	13-ಓ	14-ಔ	15-ಕ	16-ಕೆ	17-ಕಾ	18-ಕಿ	19-ಕು	20-ಕೂ
21-ಕೈ	22-ಕೌ	23-ಕೇ	24-ಕೈ	25-ಕೊ	26-ಕೌ	27-ಕಂ	28-ಖ	29-ಖ	30-ಖಾ
31-ಖಿ	32-ಖು	33-ಖೂ	34-ಖೈ	35-ಖೌ	36-ಖೇ	37-ಖೇ	38-ಖೈ	39-ಖೊ	40-ಖೌ
41-ಗ	42-ಗ	43-ಗಾ	44-ಗಿ	45-ಗು	46-ಗೂ	47-ಗೈ	48-ಗೌ	49-ಗೇ	50-ಗೈ
51-ಗೊ	52-ಗೌ	53-ಘ	54-ಘ	55-ಘಾ	56-ಘಿ	57-ಘೇ	58-ಘು	59-ಘೂ	60-ಘೈ
61-ಘೌ	62-ಘೇ	63-ಘೈ	64-ಘೊ	65-ಘೌ	66-ಘೇ	67-ಘ	68-ಘ	69-ಘ	70-ಘ
71-ಙ	72-ಙ	73-ಙಾ	74-ಙಿ	75-ಙು	76-ಙೂ	77-ಙೈ	78-ಙೌ	79-ಙೇ	80-ಙೈ
81-ಙೊ	82-ಙೌ	83-ಙೇ	84-ಙೈ	85-ಙೌ	86-ಙೇ	87-ಙೇ	88-ಙೈ	89-ಙೊ	90-ಙೌ
91-ಞ	92-ಞ	93-ಞಾ	94-ಞಿ	95-ಞು	96-ಞೂ	97-ಞೈ	98-ಞೌ	99-ಞೇ	100-ಞೈ
101-ಞೊ	102-ಞೌ	103-ಜ	104-ಜ	105-ಜಾ	106-ಜಿ	107-ಜು	108-ಜೂ	109-ಜೈ	110-ಜೌ
111-ಜೇ	112-ಜೈ	113-ಜೊ	114-ಜೌ	115-ಝ	116-ಝ	117-ಝಾ	118-ಝಿ	119-ಝು	120-ಝೂ
121-ಝು	122-ಝೌ	123-ಝು	124-ಝು	125-ಝು	126-ಝು	127-ಞ	128-ಞ	129-ಞಾ	130-ಞ
131-ಞೇ	132-ಞಾ	133-ಞಾ	134-ಞು	135-ಞು	136-ಞ	137-ಞೇ	138-ಞೈ	139-ಞೌ	140-ಞೌ
141-ಟ	142-ಟ	143-ಟಾ	144-ಟಿ	145-ಟು	146-ಟೂ	147-ಟೈ	148-ಟೌ	149-ಟೇ	150-ಟೈ
151-ಟೈ	152-ಟೊ	153-ಟೌ	154-ಠ	155-ಠ	156-ಠಾ	157-ಠಿ	158-ಠು	159-ಠೂ	160-ಠೈ
161-ಠೌ	162-ಠೇ	163-ಠೇ	164-ಠೈ	165-ಠೊ	166-ಠೌ	167-ಠೇ	168-ಠೈ	169-ಠೊ	170-ಠೌ
171-ಠೂ	172-ಠು	173-ಠೂ	174-ಠು	175-ಠು	176-ಠು	177-ಠು	178-ಠು	179-ಠು	180-ಠು
181-ಠಿ	182-ಠು	183-ಠೂ	184-ಠು	185-ಠೂ	186-ಠು	187-ಠೇ	188-ಠೈ	189-ಠೌ	190-ಠೌ
191-ಠ	192-ಠ	193-ಠಾ	194-ಠಿ	195-ಠು	196-ಠೂ	197-ಠು	198-ಠು	199-ಠ	200-ಠೇ
201-ಠೈ	202-ಠೊ	203-ಠೌ	204-ಠ	205-ಠ	206-ಠಾ	207-ಠಿ	208-ಠು	209-ಠೂ	210-ಠು
211-ಠು	212-ಠ	213-ಠೇ	214-ಠೈ	215-ಠೊ	216-ಠೌ	217-ಠೇ	218-ಠು	219-ಠಾ	220-ಠಿ
221-ಠು	222-ಠೂ	223-ಠು	224-ಠು	225-ಠು	226-ಠು	227-ಠು	228-ಠು	229-ಠು	230-ಠ
231-ಠ	232-ಠಾ	233-ಠಿ	234-ಠು	235-ಠೂ	236-ಠು	237-ಠು	238-ಠ	239-ಠೇ	240-ಠೈ
241-ಠೊ	242-ಠೌ	243-ಠೇ	244-ಠು	245-ಠಾ	246-ಠಿ	247-ಠು	248-ಠೂ	249-ಠು	250-ಠು
251-ಠ	252-ಠು	253-ಠೂ	254-ಠು	255-ಠ	256-ಠ	257-ಠಾ	258-ಠಿ	259-ಠು	260-ಠೂ
261-ಠು	262-ಠು	263-ಠ	264-ಠೇ	265-ಠೈ	266-ಠೊ	267-ಠೌ	268-ಠೇ	269-ಠು	270-ಠಾ
271-ಠಿ	272-ಠು	273-ಠೂ	274-ಠು	275-ಠೂ	276-ಠು	277-ಠು	278-ಠು	279-ಠು	280-ಠ
281-ಠ	282-ಠಾ	283-ಠಿ	284-ಠು	285-ಠೂ	286-ಠು	287-ಠು	288-ಠು	289-ಠು	290-ಠು
291-ಠ	292-ಠ	293-ಠ	294-ಠಾ	295-ಠಿ	296-ಠು	297-ಠೂ	298-ಠು	299-ಠು	300-ಠ
301-ಠೇ	302-ಠೈ	303-ಠೊ	304-ಠೌ	305-ಠೇ	306-ಠು	307-ಠಾ	308-ಠಿ	309-ಠು	310-ಠೂ
311-ಠು	312-ಠು	313-ಠು	314-ಠೇ	315-ಠೈ	316-ಠೊ	317-ಠೌ	318-ಠೇ	319-ಠು	320-ಠಾ
321-ಠಿ	322-ಠು	323-ಠೂ	324-ಠು	325-ಠೂ	326-ಠು	327-ಠು	328-ಠೂ	329-ಠು	330-ಠು
331-ಠು	332-ಠಾ	333-ಠಿ	334-ಠು	335-ಠೂ	336-ಠು	337-ಠು	338-ಠು	339-ಠು	340-ಠೂ
341-ಠು	342-ಠ	343-ಠ	344-ಠಾ	345-ಠಿ	346-ಠು	347-ಠೂ	348-ಠು	349-ಠು	350-ಠ
351-ಠೇ	352-ಠೈ	353-ಠೊ	354-ಠೌ	355-ಠೇ	356-ಠು	357-ಠಾ	358-ಠಿ	359-ಠು	360-ಠೂ
361-ಠ	362-ಠೇ	363-ಠು	364-ಠು	365-ಠು	366-ಠ	367-ಠ	368-ಠಾ	369-ಠಿ	370-ಠು
371-ಠೂ	372-ಠು	373-ಠು	374-ಠ	375-ಠೇ	376-ಠೈ	377-ಠೊ	378-ಠೌ	379-ಠೇ	380-ಠ
381-ಠಾ	382-ಠಿ	383-ಠು	384-ಠೂ	385-ಠು	386-ಠು	387-ಠ	388-ಠೇ	389-ಠೈ	390-ಠೊ
391-ಠೌ	392-ಠೇ	393-ಠು	394-ಠ	395-ಠ	396-ಠೇ	397-ಠು	398-ಠೂ	399-ಠು	400-ಠು
401-ಠ	402-ಠೇ	403-ಠು	404-ಠು	405-ಠು	406-ಠ	407-ಠ	408-ಠಾ	409-ಠಿ	410-ಠು
411-ಠೂ	412-ಠು	413-ಠು	414-ಠ	415-ಠೇ	416-ಠೈ	417-ಠೊ	418-ಠೌ	419-ಠೇ	420-ಠ
421-ಠಾ	422-ಠಿ	423-ಠಿ	424-ಠು	425-ಠೂ	426-ಠು	427-ಠು	428-ಠ	429-ಠೇ	430-ಠೈ
431-ಠೂ	432-ಠೌ	433-ಠಂ	434-ಠ	435-ಠ	436-ಠಾ	437-ಠಿ	438-ಠು	439-ಠೂ	440-ಠು
441-ಠು	442-ಠ	443-ಠು	444-ಠೂ	445-ಠೌ	446-ಠ	447-ಠ	448-ಠಾ	449-ಠಿ	450-ಠು
451-ಠೂ	452-ಠು	453-ಠು	454-ಠ	455-ಠು	456-ಠೂ	457-ಠೌ	458-ಠ	459-ಠ	460-ಠಾ
461-ಠಿ	462-ಠು	463-ಠೂ	464-ಠು	465-ಠು	466-ಠ	467-ಠು	468-ಠೂ	469-ಠೌ	470-ಠ
471-ಠ	472-ಠಾ	473-ಠಿ	474-ಠಿ	475-ಠು	476-ಠು	477-ಠು	478-ಠು	479-ಠ	480-ಠೇ
481-ಠೈ	482-ಠೊ	483-ಠೌ	484-ಠ	485-ಠ	486-ಠ	487-ಠ	488-ಠ	489-ಠ	490-ಠ
491-ಠ	492-ಠ	493-ಠ							

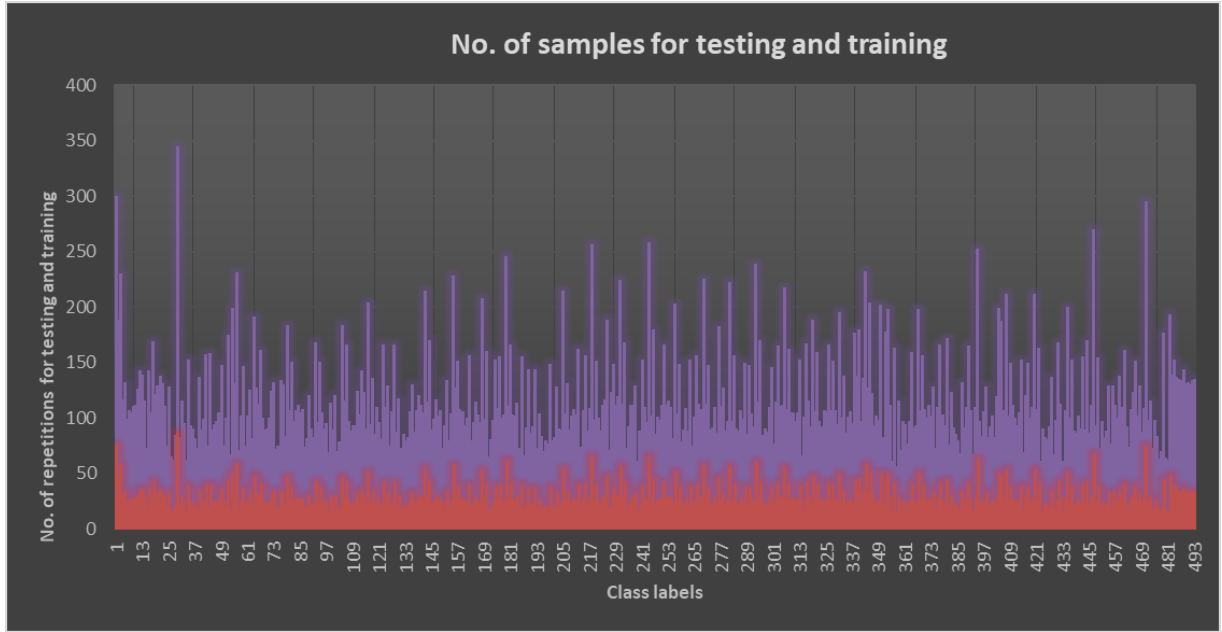


Figure 2: No. of Samples in data – Class labels Vs no. of repetitions

Table 2: Details dataset distribution

Total number of Classes	Total Number of Samples		Approximate no. of samples in each class	
	Training data	Testing data	Training data	Testing data
493	58014	14504	55-345	15-85

Table 1 provides the distribution of sample data across different classes for both training and testing purposes. On the other hand, Table 2 presents a summary of the training and testing datasets for the 493 classes, indicating the total number of instances available for each class in both the training and testing sets. Figure 2 visually illustrates the distribution of samples across different class labels for both testing and training datasets.

The methodology employed in this study involves utilizing Resnet50 for the classification of handwritten Kannada characters, which exhibits an architecture as depicted in Figure 3. The Resnet50 model consists of four hidden layers, a fully connected layer, and finally an output layer. To begin, the input layer of Resnet50 receives unprocessed input images having dimensions of approximately 224x224 pixels. The architecture utilizes residual blocks, where the block comprises multiple convolutional layers, followed by batch normalization and ReLU activation.

These residual blocks facilitate the extraction of essential features and enhance the representation learning process. The subsequent blocks in the Resnet50 architecture progressively refine the learned representations and extract more intricate features from the input images. The final set of residual blocks in the architecture captures the most abstract and high-level features present in the input images, contributing to improved recognition performance.

Following the residual blocks, an average pooling layer is applied to the feature maps obtained. This pooling operation computes the average value for each feature map, further abstracting the features. Finally, the output layer of the Resnet50 model maps the abstracted features to the 493 classes of handwritten Kannada characters.

In Table 2, detailed information regarding the datasets that have been used for training and testing the Resnet50 model on Kannada handwritten characters is provided

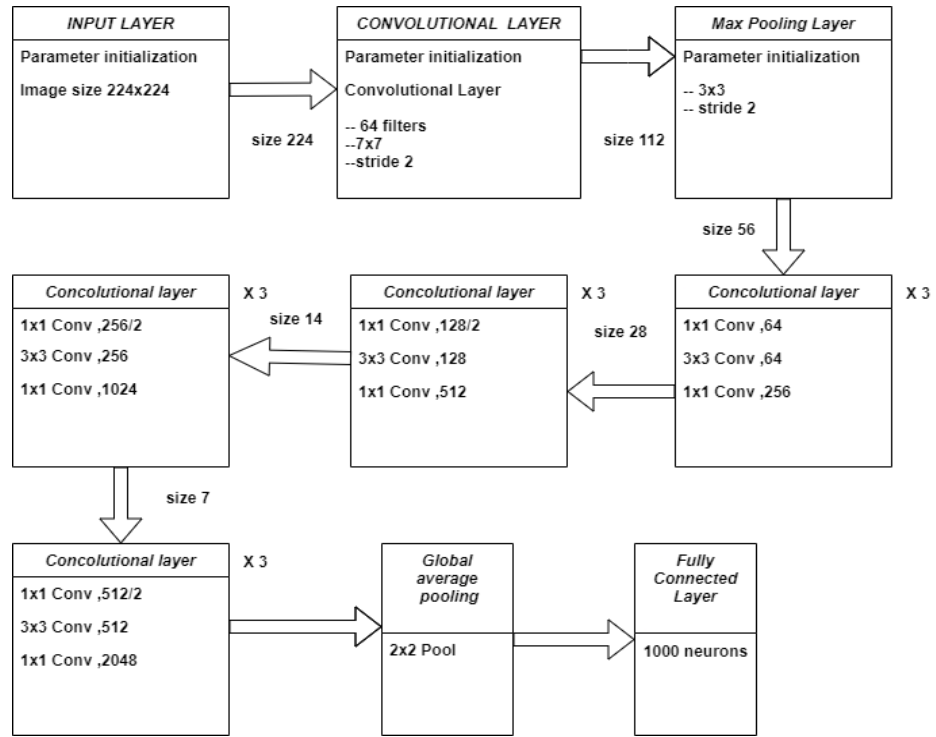


Figure 3: Architecture of Resnet50

II. EXPERIMENTAL ANALYSIS

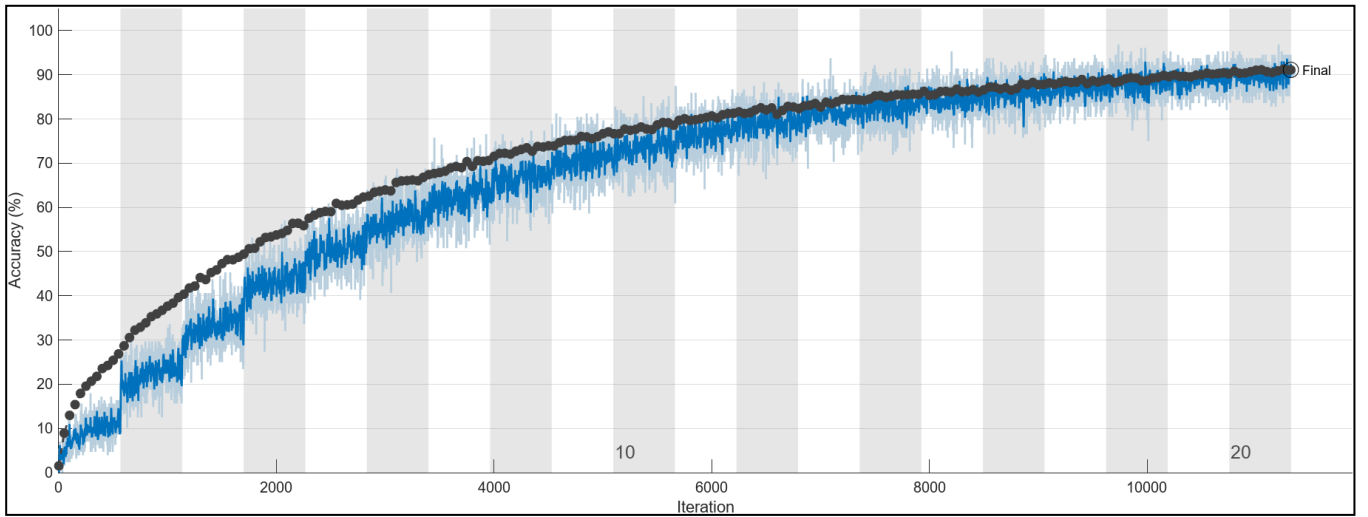


Figure 4: Evaluation of ResNet50 Performance: Relationship between Training Epochs and Accuracy

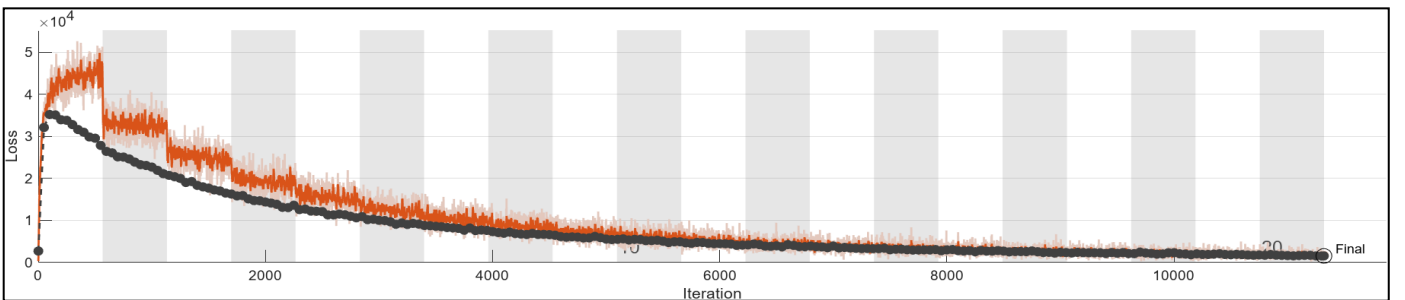


Figure 5: Evaluation of ResNet50 Performance: Relationship between Training Epochs and Loss

classes representing the Kannada alphabetical set. The training dataset consisted of 58,014 instances, while the test dataset contained 14,504 instances. Table 2 provides detailed statistics regarding the distribution of instances across each class.

For the classification process, we utilized ResNet50, and the training procedure was carried out over a span of 20 epochs. During each epoch, we performed 566 iterations to classify the test instances. The experimental analysis includes two key graphical representations. Figure 4 showcases the accuracy versus the number of epochs, highlighting the performance improvement over iterations. Notably, the validation accuracy achieved in the experiments is recorded at 91.11%. Figure 5 depicts the loss versus the number of epochs, providing insights into the convergence and optimization process of the model.

These experimental results demonstrate the effectiveness and reliability of the proposed methodology in recognizing handwritten Kannada characters. The use of synthetically generated datasets and the integration of Resnet50 contribute to achieving high accuracy and robust classification performance.

CONCLUSION

The utilization of a substantial amount of training knowledge, coupled with the adoption of transfer learning, has yielded satisfactory results in recognizing a vast number of classes. With 493 classes in the Kannada character dataset, the proposed methodology has achieved an impressive accuracy rate of approximately 91.11% during validation. However, it is worth noting that further improvements can be attained by enhancing the resolution of the dataset. Additionally, increasing the number of instances within each class has the potential to boost the recognition rate even further.

The findings of this study highlight the effectiveness and potential of employing transfer learning techniques, specifically using Resnet50 architecture, for handwritten Kannada character recognition. The demonstrated accuracy rate signifies the model's capability to handle a large number of classes and capture the intricate features of the Kannada script. As future work, exploring higher-resolution datasets and expanding the dataset to include more instances per class can lead to even higher recognition rates and further improvements in the overall performance of the system. By continuously refining the dataset and exploring advancements in deep learning techniques, we can continue to enhance the accuracy and robustness of the system for real-world applications

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